

LSGI2801: Smart Cities Context, Policy and Government

A Comparative Analysis of Smart City Trajectories: Mobility, Energy, and Economy in Hong Kong and Singapore

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Abstract

This study presents a comparative analysis of smart city trajectories in Hong Kong and Singapore, focusing on three interlinked dimensions: smart mobility, smart energy, and smart economy. To systematically evaluate these urban models, this research addresses four primary objectives. First, it identifies the specific technologies and infrastructure deployed in both cities, contrasting Hong Kong's market-driven, evolutionary approach with Singapore's government-led, transformative paradigm. Second, the study quantitatively analyzes the interactions among these dimensions using spatio-temporal data visualization and Ordinary Least Squares (OLS) regression models. The empirical analysis not only evaluates the effectiveness of dynamic pricing on traffic congestion but also reveals an unexpected negative correlation between local economic prosperity and Electric Vehicle (EV) infrastructure density, driven by distinct mobility-centered mechanisms in both cities. Third, drawing on these empirical findings, the study evaluates the positive and negative impacts of these developmental paths. The findings clarify a fundamental trade-off: Hong Kong demonstrates high market adaptability but struggles with system fragmentation and spatial inequity, whereas Singapore achieves sustained system efficiency and equitable infrastructure distribution at the expense of centralized rigidity and restricted individual mobility choices. Finally, the paper puts forward practical recommendations to address these ongoing challenges. It concludes that a hybrid governance model—combining government-led equity and open data standards with market-driven local innovation—provides the most sustainable framework for future smart city development.

Keywords

Smart Cities; Hong Kong; Singapore; Smart Mobility; Smart Energy; Smart Economy

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1 Introduction

1.1 Background

The concept of smart cities can be traced back to the early 21st century, evolving from IBM's "Smarter Planet" initiative (Rubin, 2008). Rudolf Giffinger et al. systematically proposed six core dimensions of a smart city: smart economy, smart mobility, smart environment, smart people, smart living and smart governance (Giffinger et al., 2007). According to their definition, a city can be defined as a smart city only if it performs well in all six dimensions.

This paper selects three of these dimensions for discussion. The first is the smart economy, which encompasses innovation, entrepreneurship, productivity, flexibility in the labor market, and the degree of integration into the international market. The second is smart mobility, which encompasses not only modern transport infrastructure but also sustainable transport systems, the accessibility of ICT infrastructure, and safe and efficient transport. The third is the smart environment, which involves the protection of natural resources, pollution control, sustainable resource management, and an emphasis on environmental protection.

In recent years, both Hong Kong and Singapore have actively promoting the development of smart cities. They are similar in many ways: both are small and highly developed city-states with strong economies and effective public governance (Kratoska, 2006). Due to these similarities, it is easier to control variables when making comparisons. Therefore, the two cities are often selected as typical cases for a comparative analysis of smart city development (Ang-Tan & Ang, 2022).

1.2 Literature Review

With the deepening of research, scholars have expanded the concept of smart cities from early technology initiatives to multidimensional evaluation frameworks. In particular, the six-dimension framework proposed by Giffinger et al. has been widely adopted to organize comparative assessments across economy, mobility, environment, people, living, and governance (Giffinger et al., 2007). Within urban planning and transport studies, smart mobility theory has become a common lens for transportation network design, emphasizing not only advanced infrastructure but also sustainability, ICT accessibility, and safety.

More recently, the integration of smart energy and the smart economy has become a research focus. The deployment of EV charging infrastructure is increasingly recognized as a critical intersection across these dimensions: EV networks provide the physical foundation that sustains smart mobility, while their spatial planning is shaped by economic vitality and local property or governance models.

1.3 Restatement of the Problem

Our main challenge is to examine the development of smart cities in Hong Kong and Singapore, with a focus on three aspects: smart mobility, smart economy, and smart environment. Specifically, our research objectives are as follows.

- To identify the technologies, infrastructure, and solutions that have been developed to achieve smart city applications.
- To analyze the interactions among the three dimensions mentioned above.
- To evaluate the positive and negative impacts arising from smart city development.
- To put forward practical recommendations for the future development of smart cities.

1.4 Our work

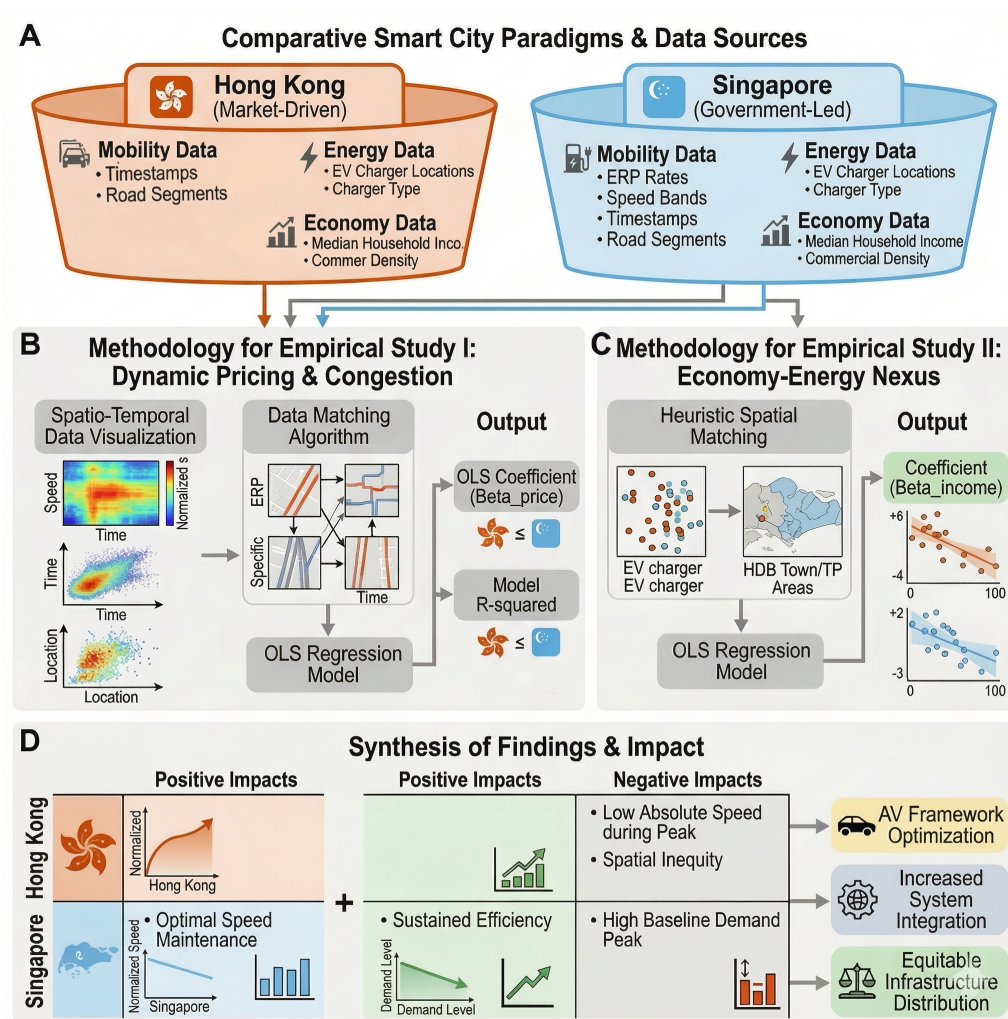


Figure 1: Work Flow

2 Identification of Smart City Technologies and Infrastructure

To systematically compare the smart city paradigms of Hong Kong and Singapore, it is imperative to examine the specific technologies and infrastructure deployed within these high-density Asian metropolises. Although both regions share similarities such as land scarcity, strong economies, and robust public governance (Wang, 2018), their distinct urban governance models have fostered divergent technological trajectories. This section identifies core technologies across three primary dimensions: smart mobility, smart energy, and the smart economy.

2.1 Smart Mobility: Efficiency and Autonomy

2.1.1 Strategic Management Systems (SIGMA vs. LTMP 2040)

Hong Kong and Singapore have adopted fundamentally different strategic management systems to address urban mobility. Hong Kong pursues a market-driven strategy that prioritizes the optimization of existing transportation facilities through Intelligent Transport Systems (ITS) (Transport Department, The Government of the Hong Kong Special Administrative Region, 2021). Within the “SIGMA” vision, Hong Kong aims to leverage real-time data integration and intelligent traffic law enforcement to create a safe, informative, green, convenient (mobile), and accessible transportation network (ERTICO - ITS Europe, 2019; Transport Department, The Government of the Hong Kong Special Administrative Region, 2021). This approach improves commuting efficiency without drastically altering existing traffic modes.

In contrast, Singapore employs a government-led approach to fundamentally reshape travel habits. Guidanced by the “Smart Nation” initiative (GovTech Singapore, 2023) and the Land Transport Master Plan 2040 (LTMP 2040) (Land Transport Authority, 2019), Singapore enforces a philosophy of “car-lite to mitigate the dependency on private vehicles (Centre for Liveable Cities, 2017; GovTech Singapore, 2023). Through quantitative planning frameworks such as the “45-minute city” and the “20-minute town” (Land Transport Authority, 2019), Singapore uses regulatory mechanisms and infrastructure planning to prioritize public and shared transportation.

As illustrated in Figures 2 and 3, Singapore’s “car-lite” strategy reduced household car ownership from 42.1% (2013 peak) to 36% (2023) while increasing daily public transport ridership from 4.9 million (2020) to more than 7.1 million (2025), empirically validating the success of LTMP 2040.

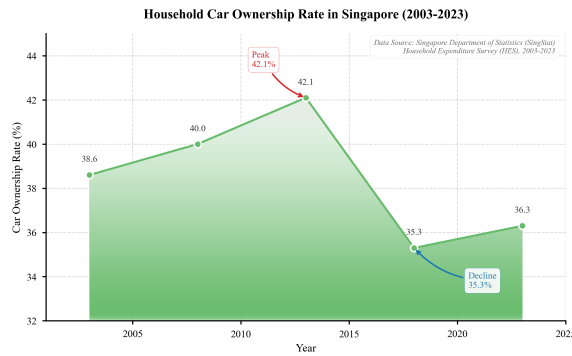


Figure 2: Household Car Ownership Rate in Singapore (2003-2023).

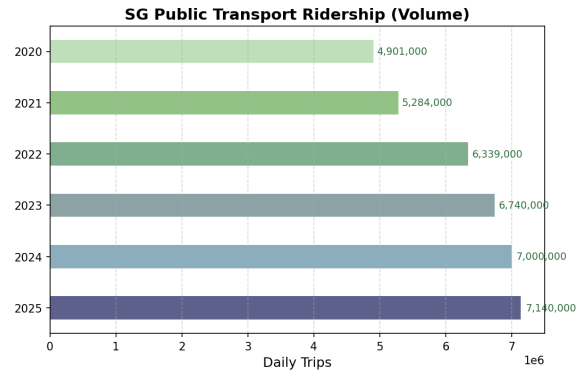


Figure 3: Daily Public Transport Ridership Volume (2020-2025).

2.1.2 Autonomous Vehicle (AV) Infrastructure

The deployment of Autonomous Vehicles (AVs) further highlights the divergence between the two cities, largely driven by their physical environments (Murshed et al., 2018). Singapore, a global pioneer in AV technology, is heavily dependent on Vehicle to Everything (V2X) coordination within standardized testing environments such as the CETRAN test center and the One-North district (Centre of Excellence for Testing & Research of Autonomous Vehicles @ NTU, n.d.). Singapore's AV infrastructure is deeply integrated with GPS and road communication units, facilitating efficient system-level scheduling suitable for public transport and port logistics (Land Transport Authority, 2023). As highlighted by Schrag (2020), the successful macroscopic deployment of shared autonomous vehicles is highly dependent on such coordinated, system-level dispatching architectures (Schrag, 2020).

In contrast, Hong Kong faces the unique challenge of high-density urban canyons, which frequently cause satellite signal blockages (Yang et al., 2020). Consequently, Hong Kong's AV development emphasizes vehicle-centric intelligence (Applied Science and Technology Research Institute Hong Kong, 2021). Using technologies such as LiDAR-based Simultaneous Localization and Mapping (SLAM) and edge computing, Hong Kong enables AVs to operate independently in complex, dynamic environments with limited external signal support (Jiang et al., 2025).

Table 1 summarizes the divergent AV infrastructure trajectories between the two cities, driven by their distinct physical environments.

Table 1: Comparison of Autonomous Vehicle (AV) Infrastructure and Technologies

Feature	Singapore	Hong Kong
Primary Technology	Vehicle to Everything (V2X)	LiDAR based SLAM and edge computing
Navigation Reliance	GPS and roadside communication units	Vehicle centric independent operation
Testing Environment	Standardized centers (e.g., CE-TRAN)	High density urban canyons
System Architecture	System level efficient scheduling	Independent vehicle intelligence

2.2 Smart Energy: Sustainable Infrastructure and Grid Intelligence

2.2.1 EV Charging Networks

A critical component of the dimension of smart environment within urban planning is sustainable resource management and pollution control (Ahvenniemi et al., 2017). Consequently, the transition to sustainable urban transportation requires the development of robust and accessible Electric Vehicle (EV) charging networks (Hardman et al., 2018). Integrating the findings of Zhuge et al. (2019) regarding the spatial-temporal dynamics of EV charging, it is evident that high-density cities must strategically optimize charging station locations to prevent local power grid overloads and alleviate driver range anxiety (Zhuge et al., 2019).

To achieve these optimization goals, different high-density cities adopt distinct governance models tailored to their urban landscapes. For example, Singapore integrates the charging infrastructure of electric vehicles into its centralized government-planned public housing (HDB) and commercial hubs to support its broader “car-lite” and green transit objectives (Land Transport Authority, 2021). In contrast, Hong Kong relies predominantly on market-driven installations within private commercial properties, a strategy that aligns closely with its pragmatic, market-oriented adaptation approach to urban sustainability (Environment Bureau, 2021).

2.2.2 Smart Grid and Renewable Integration

Smart grid technologies are the foundation for a smart environment. They help protect natural resources and support sustainable energy management (Ahvenniemi et al., 2017). With advanced metering infrastructure (AMI) and renewable energy sources, cities can improve environmental protection (Mohassel et al., 2014). The relationship between smart mobility and the smart environment can be clearly seen here. Smart Well-managed networks are needed to power electric transport while keeping urban energy supplies stable (Zhuge et al., 2021).

2.3 Smart Economy: Digital Ecosystems and Payments

2.3.1 Evolutionary vs. Disruptive Payment Technologies

A smart economy relies heavily on technological innovation (Caragliu et al., 2011). Hong Kong and Singapore adopt very different approaches to smart payment systems. Hong Kong shows a gradual evolution, whereas Singapore shows a rapid transformation.

In Hong Kong, the Octopus card heavily influences the mobile payment landscape along with other digital wallets. This system uses a closed-loop stored-value design based on FeliCa technology (Chau & Poon, 2003). It enables fast and offline transactions. However, this market dominance hinders the adoption of new technologies. The transition to open payment systems remains slow. Currently, different providers offer fragmented QR code and EMV card solutions.

In contrast, Singapore follows a government-led approach to smart payments. The SimplyGo initiative transitioned the city to an Account-Based Ticketing (ABT) system. The system processes fares on central servers instead of physical cards. This shift creates a fully open payment environment. Commuters can easily pay with standard contactless bank cards (Aw et al., 2021).

2.3.2 Digital Infrastructure for Innovation

The design of payment systems directly affects the digital infrastructure of a city and the development of Mobility-as-a-Service (MaaS) (Kamargianni et al., 2016). Hong Kong has a divided payment system. This fragmentation makes it difficult to build a single MaaS platform that connects different transport modes (Jittrapirom et al., 2017). In contrast, Singapore uses an Account-Based ticketing system (ABT). This ABT system allows the Land Transport Authority to collect complete and accurate traffic data (Aw et al., 2021). Centralized data strongly supports improvements in urban transport. It also encourages innovation in the wider smart economy (Lim et al., 2018).

2.4 Section Summary: Comparative Technological Paradigms

In summary, there is a gap between the smart city paradigms of Hong Kong and Singapore. Hong Kong follows a market-driven approach that gradually develops. Focuses on vehicle technology and improves existing systems, such as the Octopus card and intelligent transport systems. The city aims to adapt practically to its highly crowded urban environment. Singapore, however, uses a government-led model that brings sudden and deep changes. With centralized projects such as SimplyGo and well-coordinated V2X systems, Singapore actively rebuilds its urban transport systems. This helps the city achieve general reform and better data integration.

3 Methodology and Data Framework

3.1 Research Design

This study employs a comparative, quantitative research design to evaluate the smart city trajectories of Hong Kong and Singapore. The empirical analysis is divided into two primary dimensions. The first is a temporal and spatial evaluation of dynamic pricing strategies to determine their effectiveness on traffic efficiency. The second involves a spatial cross-sectional analysis to investigate the nexus of the smart economy and smart energy, specifically examining how local economic prosperity influences the deployment of sustainable Electric Vehicle (EV) infrastructure.

3.2 Data Sources

All datasets are obtained from official government open-data portals. Smart mobility data include Hong Kong time-varying tunnel tolls and corresponding traffic speeds (km/h), and Singapore ERP rates (SGD) with traffic speed bands (1–5). Smart energy/economy data include EV-charger locations and census-level household income (“high-income” defined as $> 30,000$ HKD in Hong Kong and $\geq 10,000$ SGD in Singapore).

3.3 Statistical Models

The core analytical framework relies on Ordinary Least Squares (OLS) regression models to quantify the relationships between both empirical studies. Before modeling, spatiotemporal data visualization techniques, including heat maps and speed distribution charts, were utilized to identify preliminary localized patterns.

For Empirical Study I, data matching algorithms were employed to align timestamps and specific road segments (e.g. CHT, EHC, WHC in Hong Kong) with their respective pricing records before fitting the regressions. The traffic efficiency models are defined as follows.

- **Hong Kong:** $\text{Speed_kmh} = \beta_0 + \beta_1 \cdot \text{Current_Toll} + \epsilon$
- **Singapore:** $\text{SpeedBand} = \alpha + \gamma_1 \times \text{Avg_ERP_Rate} + \mu$

For Empirical Study II, the data were geographically aggregated by District for Hong Kong and by Planning Area for Singapore. A heuristic spatial matching approach was specifically developed for the Singapore dataset to map address-level EV charger data (street and building names) to broader planning areas. The resulting model evaluates spatial equity as follows:

- **EV Infrastructure Model:** $\text{Charger_Count} = \beta_0 + \beta_1 \cdot \text{High_Income_Ratio} + \epsilon$

4 Empirical Study I: Effectiveness of Smart Mobility Solutions

4.1 Scenario and Variable Selection

This section evaluates dynamic pricing strategies in Hong Kong and Singapore. We analyze their impact on traffic efficiency. We use two sets of variables. For Hong Kong, the independent variable is the Time-varying Toll (in HKD), and the dependent variable is the Traffic Speed (in km/h). For Singapore, the independent variable is the Electronic Road Pricing (ERP) Rate (in SGD), and the dependent variable is the Traffic Speed Band (scale 1 to 5). We extract all data from open government data portals.

4.2 Spatio-Temporal Data Visualization

Before running the regression, we visualize the temporal and spatial distribution of the pricing systems.

The Singapore ERP System: The ERP rates heatmap (Figure 4) clearly demonstrates a highly targeted dynamic pricing mechanism. The government applies the highest ERP rates strictly during the morning and evening peak hours.

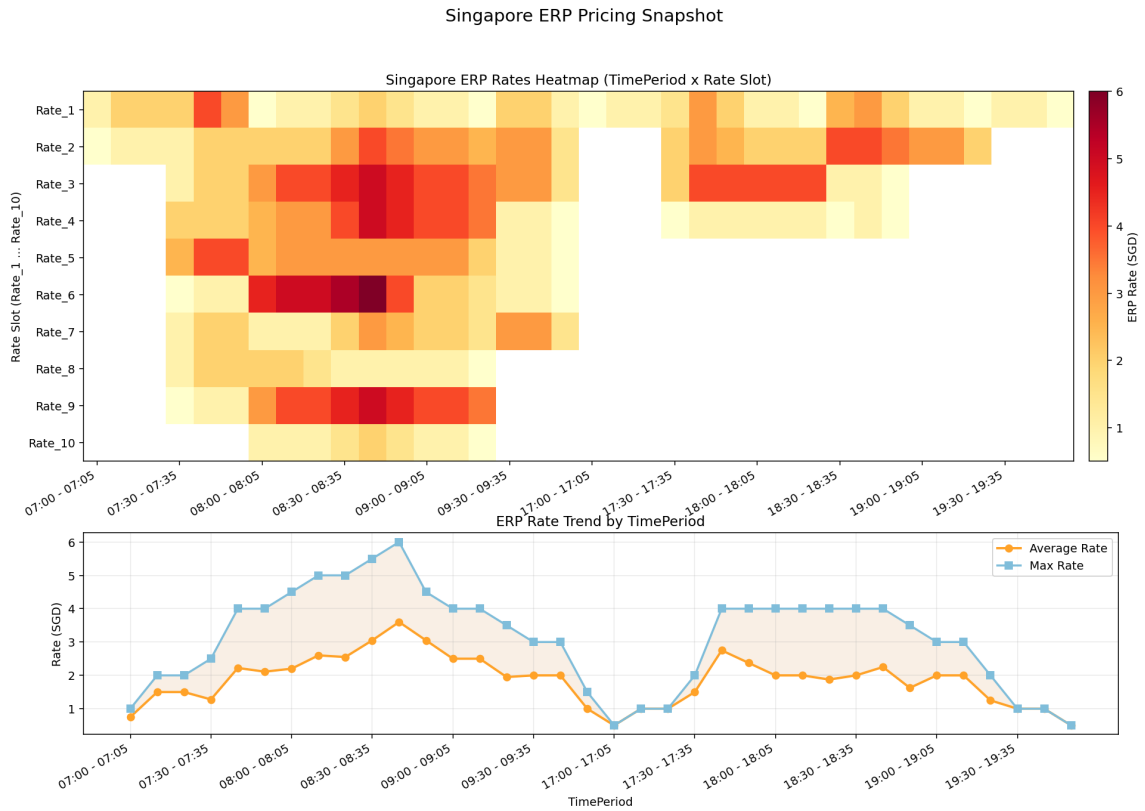


Figure 4: Spatiotemporal Heatmap of Singapore ERP Rates

The speed heat distribution chart (Figure 5) proves the overall success of this system. Most major expressways maintain a high proportion of Speed Band 4 and 5, indicating smooth traffic flow.

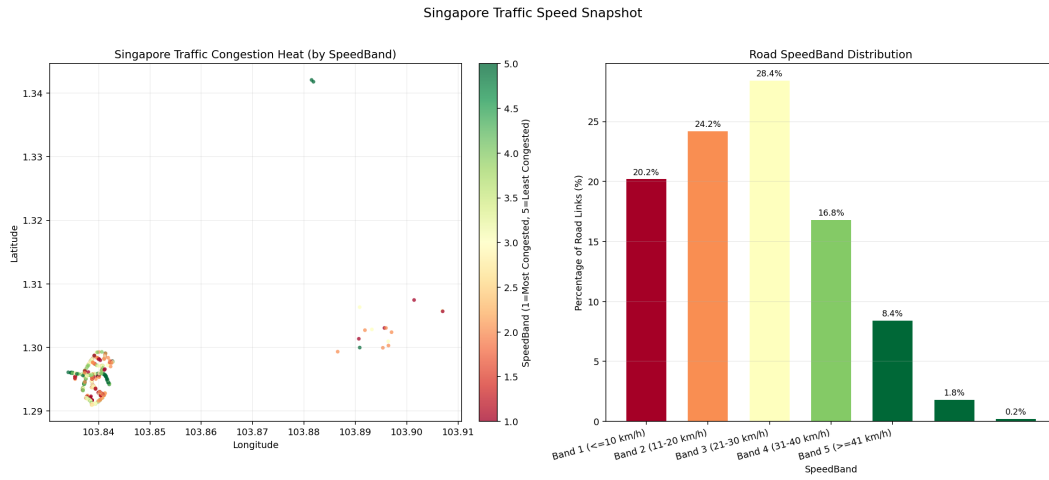


Figure 5: Distribution of Traffic Speed Bands Across Major Expressways in Singapore

The Hong Kong Toll System: The subplots for Hong Kong (Figure 6) display data for the three cross-harbour tunnels (CHT, EHC, and WHC). The charts reveal a strong temporal overlap between peak pricing and peak congestion. The toll rates rise sharply during the morning and evening rush hours. At the same time, the traffic speeds drop significantly. This creates a clear inverse relationship. Notably, the WHC imposes the highest peak toll at 60 HKD. Consequently, the WHC maintains a higher and more stable traffic speed during peak hours compared to the CHT and EHC.

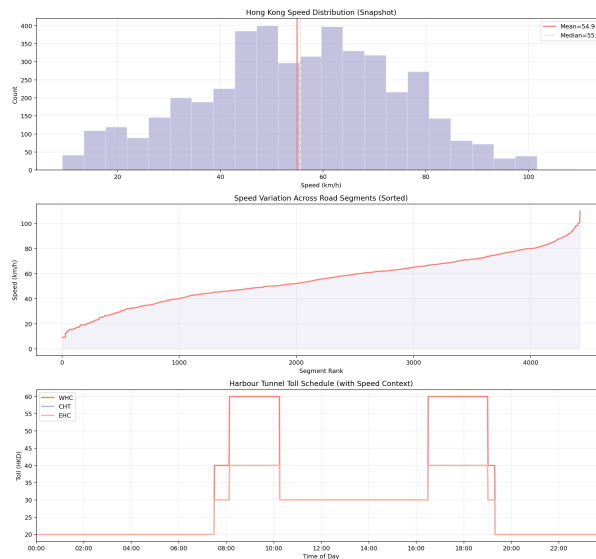


Figure 6: Hong Kong Speed Distribution and Harbour Tunnel Toll Schedule

4.3 Regression Algorithm and Model Setup

To quantify the relationship between dynamic pricing and traffic congestion, we process the raw datasets and apply Ordinary Least Squares (OLS) regression models. The data matching and modeling process is detailed in Algorithm 1.

Algorithm 1 Dynamic Pricing and Traffic Speed Regression Analysis

Require: HK Speed Data, HK Pricing Data, SG Speed Data, SG ERP Data

Ensure: OLS Regression Models for HK and SG

- 1: **Part 1: Hong Kong Data Processing**
 - 2: Filter HK Speed Data for specific Tunnel Segments (CHT, EHC, WHC)
 - 3: **for** each record in HK Speed Data **do**
 - 4: Match Time and Tunnel_Name with HK Pricing Data
 - 5: Assign corresponding Current_Toll
 - 6: **end for**
 - 7: Remove records with missing values (Speed_kmh or Current_Toll)
 - 8: Fit Model: $\text{Speed_kmh} = \beta_0 + \beta_1 \cdot \text{Current_Toll} + \epsilon$
 - 9: **Part 2: Singapore Data Processing**
 - 10: Aggregate SG ERP Data by RoadName to calculate Avg_ERP_Rate
 - 11: Inner Join SG Speed Data and Aggregated ERP Data on RoadName
 - 12: Remove records with missing values
 - 13: Fit Model: $\text{SpeedBand} = \alpha + \gamma_1 \times \text{Avg_ERP_Rate} + \mu$
 - 14: **return** HK Model Summary, SG Model Summary
-

4.4 Regression Results and Analysis

We perform a regression analysis to quantify these relationships. The scatter plots (Figure 7) show different patterns for the two cities.

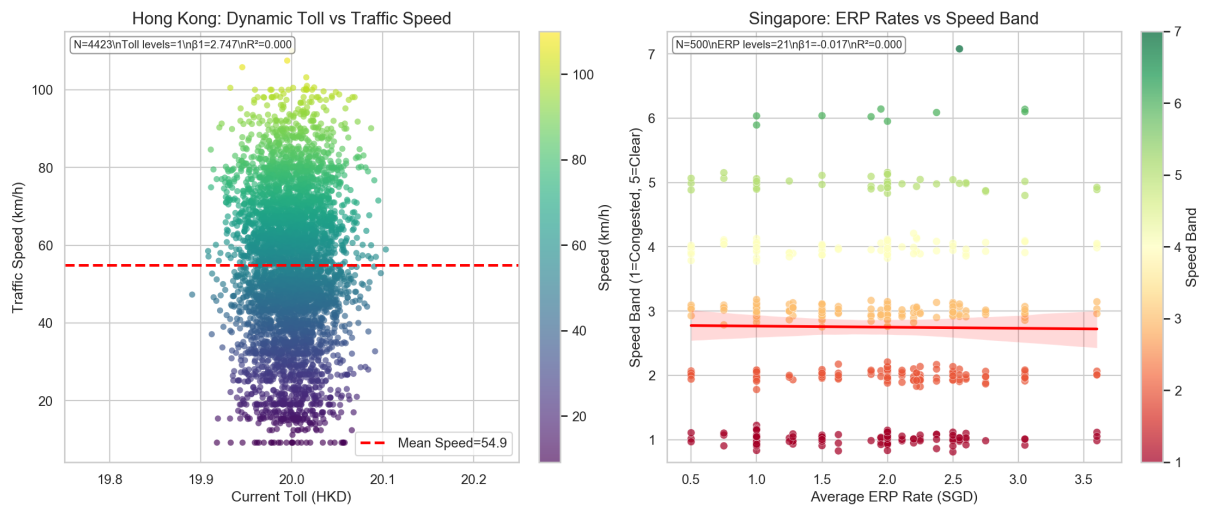


Figure 7: OLS Regression Results: Dynamic Pricing vs. Traffic Speed in HK and SG

Hong Kong Findings: The regression line for Hong Kong shows a negative correlation. Higher toll rates correlate with lower traffic speeds. This highlights a classic case of endogeneity in traffic economics. The government imposes high tolls specifically because the baseline traffic volume is overwhelming during rush hours. The pricing strategy mitigates the severity of the congestion, but it cannot entirely eliminate the peak-hour volume effect. Therefore, absolute speeds remain lowest when tolls are highest.

Singapore Findings: The regression line for Singapore shows a much flatter, more stable relationship. Even as Average ERP Rates increase, the Speed Band does not drop drastically. The data points remain clustered around acceptable speed levels. This indicates a highly responsive and effective system. The localized, high-frequency rate adjustments successfully control vehicle demand. Consequently, the ERP system maintains stable traffic speeds even under high pricing conditions.

4.5 Comparative Synthesis

The data reveals different levels of effectiveness. Singapore relies on a highly granular, location-specific ERP system. This system successfully maintains optimal traffic speeds. Hong Kong relies on broader time-varying tolls at isolated bottlenecks. While the Hong Kong model penalizes peak-hour travel, it lacks the spatial flexibility to maintain high travel speeds during absolute peak demand.

5.2 Methodology and Algorithm Design

To quantitatively evaluate the relationship between local economic prosperity and EV infrastructure density, we propose an Ordinary Least Squares (OLS) regression model. The independent variable is the High-Income Household Ratio (households earning $\geq 30,000$ HKD in Hong Kong, and $\geq 10,000$ SGD in Singapore), and the dependent variable is the number of EV chargers per district/planning area.

The data processing, spatial mapping, and regression modeling are formalized in Algorithm 2. For Singapore, a heuristic spatial matching approach was employed to map address-level charger data to broader planning areas.

Algorithm 2 Spatiotemporal Matching and Economic-Energy Regression

Require: EV Charger Datasets (C_{HK}, C_{SG}), Income Datasets (I_{HK}, I_{SG})

Ensure: OLS Regression Models (M_{HK}, M_{SG}), Comparative Visualization (V)

- 1: **Part 1: Hong Kong Data Processing**
 - 2: $C'_{HK} \leftarrow$ Group C_{HK} by District to calculate Charger_Count
 - 3: $I'_{HK} \leftarrow$ Filter I_{HK} for Year 2024 and calculate High_Income_Ratio
 - 4: $\text{Data}_{HK} \leftarrow$ Inner Join C'_{HK} and I'_{HK} on District
 - 5: $M_{HK} \leftarrow$ Fit OLS ($\text{Data}_{HK}.\text{Charger_Count} \sim \text{Data}_{HK}.\text{High_Income_Ratio}$)
 - 6: **Part 2: Singapore Data Processing**
 - 7: $I'_{SG} \leftarrow$ Extract Planning Areas and calculate High_Income_Ratio
 - 8: $C'_{SG} \leftarrow$ Map C_{SG} street/building names to I'_{SG} Planning Areas
 - 9: $\text{Data}_{SG} \leftarrow$ Inner Join C'_{SG} and I'_{SG} on Planning Area
 - 10: $M_{SG} \leftarrow$ Fit OLS ($\text{Data}_{SG}.\text{Charger_Count} \sim \text{Data}_{SG}.\text{High_Income_Ratio}$)
 - 11: **Part 3: Comparative Synthesis**
 - 12: $V \leftarrow$ Generate side-by-side scatter plots with regression lines
 - 13: **return** M_{HK}, M_{SG}, V
-

5.3 Regression Results and Analysis

Table 2 summarizes the OLS regression output extracted from the processed datasets. Figure 10 presents the comparative visual synthesis generated by the algorithm.

Table 2: OLS Regression Results: EV Charger Count vs. High-Income Ratio

City	Coefficient (β_1)	Constant (β_0)	R^2	P-value
Hong Kong	-116.62	108.69	0.150	0.112
Singapore	-547.04	424.53	0.109	0.087

5.4 Findings on Synergy and Conflict

The empirical results reveal an unexpected and significant phenomenon: both cities exhibit a negative correlation between local wealth and the density of the EV infrastructure. This contradicts the conventional assumption that higher-income demographics attract more sustainable infrastructure. However, integrating the dimension of smart mobility explains the underlying mechanisms, which fundamentally diverge between the two cities.

In Hong Kong, the deployment is highly market-driven. Districts with lower median incomes but immense commercial and transit density (e.g., Kwun Tong and Kwai Tsing) act as primary nodes for daily commute and smart mobility. These transport hubs attract the highest number of chargers due to the concentrated market demand. Here, mobility volume and the commercial economy synergistically drive the energy infrastructure.

In contrast, Singapore’s distribution reflects a strong government-led approach to spatial equity, tightly coupled with its mobility strategies. The highest concentrations of electric vehicle chargers are found in mass public housing (HDB) estates such as Tampines and Bedok, rather than in affluent private housing districts. These estates are designed as high-density transit nodes to support the nation’s “car-lite” mobility philosophy. By anchoring smart energy within these public mobility hubs, Singapore demonstrates its commitment to providing sustainable infrastructure as a public good, ensuring accessibility for the broader economic population.

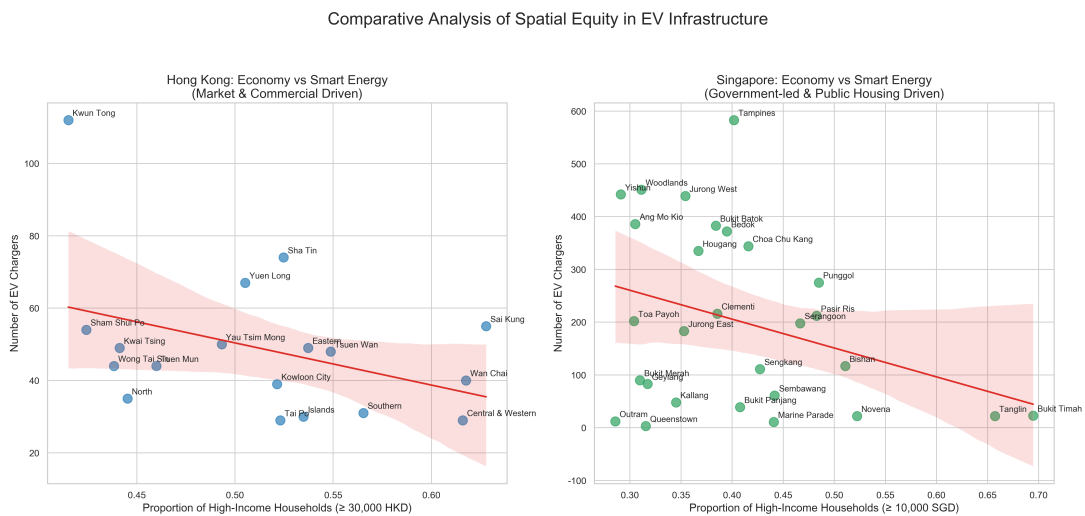


Figure 10: Comparative Analysis: The Nexus of Smart Economy and Smart Energy in Hong Kong and Singapore.

6 Evaluation of Positive and Negative Impacts

6.1 Positive Impacts

Hong Kong and Singapore have developed different paths in the building of smart cities and both have achieved considerable positive outcomes, although the results vary between the two regions.

In Singapore, the government plays an important role in urban development and this model has been effective in improving the performance of urban systems. For example, the Electronic Road Pricing system collects detailed traffic information from different areas and effectively controls the number of vehicles on the road (Parayil & Yeo, 2005). As a result, traffic speeds can remain steady even during busy hours.

Singapore also pays attention to fairness in urban space planning, which can be seen in its intelligent energy projects. The country has built a large number of charging stations for electric vehicles in public housing estates, making this public service available to more residents (Land Transport Authority, 2024). Moreover, Singapore has updated its public transport payment system to Account-Based Ticketing, known as SimplyGo (SimplyGo, 2019). This system allows flexible payment methods and collects unified data, which provides strong support for the further development of the smart economy.

Hong Kong's market-driven and pragmatic approach also presents clear advantages, particularly in adaptability. The reliance on vehicle-centric intelligence and edge computing for Autonomous Vehicles (AVs) allows them to operate independently in Hong Kong's complex, high-density urban canyons where external signals are often blocked. Economically, the market-oriented strategy has driven a rapid concentration of EV infrastructure in high-demand commercial and transit hubs, efficiently serving areas with intense daily economic activity.

6.2 Negative Impacts and Challenges

Despite these achievements, both models still encounter considerable challenges (Lam & Ma, 2019). The two cities face different obstacles in their smart city development.

Hong Kong's key difficulty is system fragmentation and insufficient spatial flexibility. The digital payment environment remains divided, with the Octopus system as the main player (Merkel, 2017).

This structure slows down the development of an integrated Mobility-as-a-Service (MaaS) platform (Kamargianni et al., 2016). For transport management, time-dependent tolls may discourage travel during peak hours, but they lack spatial flexibility and cannot maintain efficient traffic speeds under extreme demand (Cardenete & López-Cabaco, 2018). Such measures act as local bottleneck controls rather than a holistic network strategy.

In addition, the market-oriented distribution of electric vehicle chargers has led to spatial inequality (Sadeghian et al., 2021). Commercial districts often receive better services, while

residential areas may lack sufficient charging facilities (Legislative Council Secretariat, HK-SAR, 2022).

On the other hand, Singapore's dependence on centralized and systematic infrastructure creates its own challenges (Mahizhnan, 1999). The application of autonomous vehicles relies heavily on Vehicle to Everything (V2X) communication in standardized test environments. As a result, these vehicles may lack adaptability in complex and unpredictable urban situations (Lu et al., 2014). Although the "car-lite" policy effectively reduces traffic congestion, strict regulations and high costs significantly restrict individual mobility choices (Land Transport Authority, 2019). Most residents have limited options beyond public transportation (Newman & Kenworthy, 2015).

6.3 Comparative Synthesis

In summary, the comparison shows a clear trade-off between market adaptability and systemic efficiency (Kitchin, 2014). Hong Kong performs well in technologies that fit existing urban environments, such as vehicle-centered autonomous vehicles. Market forces also support fast commercial implementation. However, the city faces difficulties in system integration and spatial equity (Karvonen et al., 2019).

Singapore achieves high system efficiency, comprehensive data integration, and fair infrastructure distribution through strong top-down governance (Tan & Taeihagh, 2020). Nevertheless, this approach requires large investments in centralized infrastructure and imposes strict regulatory limits on public behavior.

7 Practical Recommendations and Conclusion

7.1 Strategic Recommendations for Hong Kong

To overcome its current challenges, Hong Kong should improve system integration and promote spatial equity.

- **Unify digital infrastructure:** The government should take a more active role in integrating the fragmented payment system. It can develop open APIs or a centralized data platform to support the establishment of a genuine MaaS system.
- **Enhance dynamic pricing:** Hong Kong may expand beyond simple tunnel tolls and adopt a more precise location-based electronic pricing scheme, similar to Singapore's ERP. This can better manage congestion across the entire urban road network.
- **Incentivize smart residential energy:** To prevent charging facilities from being concentrated only in commercial areas, the government can introduce subsidies or regulations to encourage the installation of EV chargers in residential and low-density districts.

7.2 Strategic Recommendations for Singapore

Singapore should improve the flexibility and resilience of its centralized systems.

- **Diversify AV technologies:** While V2X infrastructure is well-developed, Singapore should also support the research of vehicle-centered autonomous technologies, such as Hong Kong's LiDAR-SLAM method. This can ensure reliable operation when road communication systems are unavailable.
- **Foster market-driven innovation:** Although the state-led account-based ticketing system provides valuable data, the government should encourage private enterprises and third-party developers to create diverse applications. This approach helps avoid the dominance of state-run digital services.

7.3 Conclusion and Future Outlook

This comparative analysis indicates that no single model represents the only correct path for smart city development. Hong Kong's market-driven, evolutionary model emphasises practical adaptation and commercial viability. Singapore's government-led, transformative model focuses on system efficiency, spatial equity, and centralized management.

Findings from dynamic pricing and EV infrastructure deployment show that governance models significantly shape technological outcomes. In the future, the most sustainable smart cities will combine strengths from both approaches. A hybrid model—where the government ensures basic equity and open data standards, while market forces drive local innovation—offers the most desirable strategy for long-term urban development.

A Appendix

Experiment: Effectiveness of Smart Mobility Solutions

```
1 import pandas as pd
2 import numpy as np
3 import statsmodels.api as sm
4 import matplotlib.pyplot as plt
5 import seaborn as sns
6
7 from datetime import datetime
8 from pathlib import Path
9
10 #Set plotting style
11 sns.set_theme(style="whitegrid")
12 plt.rcParams['font.sans-serif'] = ['Arial']
13
14 #1. Read data
15 hk_speed = pd.read_csv('hk_traffic_speed_v2.csv')
16 hk_pricing = pd.read_csv('hk_harbour_pricing.csv')
17
18 known_tunnel_segments = {'1001': 'CHT', '1002': 'EHC', '1003': 'WHC'}
19 available_segment_ids = set(hk_speed['Segment_ID'].astype(str))
20 valid_tunnel_map = {k: v for k, v in known_tunnel_segments.items() if k in
    available_segment_ids}
21
22 hk_speed_filtered = pd.DataFrame()
23 hk_model_data = pd.DataFrame()
24 use_hk_fallback = len(valid_tunnel_map) == 0
25
26 if not use_hk_fallback:
27     hk_speed_filtered = hk_speed[hk_speed['Segment_ID'].astype(str).isin(
    valid_tunnel_map.keys())].copy()
28     hk_speed_filtered['Tunnel_Name'] = hk_speed_filtered['Segment_ID'].
    astype(str).map(valid_tunnel_map)
29 else:
30     hk_speed_filtered = hk_speed.copy()
31     hk_speed_filtered['Tunnel_Name'] = 'ALL_SEGMENTS'
32
33 # Time match helper
34 def get_hk_toll(time_str, tunnel_name, pricing_df, use_avg=False):
35     if pd.isna(time_str):
36         return np.nan
37     try:
38         current_time = datetime.strptime(time_str[:5], '%H:%M').time()
39         for _, row in pricing_df.iterrows():
40             start = datetime.strptime(row['Start_Time'], '%H:%M').time()
```

```

41         end = datetime.strptime(row['End_Time'], '%H:%M').time()
42         if start <= current_time <= end:
43             if use_avg:
44                 return np.mean([row['CHT_Toll'], row['EHC_Toll'], row['
WHC_Toll']])
45             if tunnel_name == 'CHT':
46                 return row['CHT_Toll']
47             if tunnel_name == 'EHC':
48                 return row['EHC_Toll']
49             if tunnel_name == 'WHC':
50                 return row['WHC_Toll']
51         except Exception:
52             pass
53         return np.nan
54
55 if not hk_speed_filtered.empty:
56     hk_speed_filtered['Current_Toll'] = hk_speed_filtered.apply(
57         lambda x: get_hk_toll(x['Time'], x['Tunnel_Name'], hk_pricing,
use_avg=use_hk_fallback), axis=1
58     )
59
60 hk_model_data = hk_speed_filtered.dropna(subset=['Speed_kmh', 'Current_Toll
'])
61 X_hk = sm.add_constant(hk_model_data['Current_Toll'])
62 y_hk = hk_model_data['Speed_kmh']
63 hk_model = sm.OLS(y_hk, X_hk).fit()
64
65 sg_speed = pd.read_csv('sg_traffic_speed.csv')
66 sg_erp = pd.read_csv('sg_erp_rates.csv')
67
68 sg_erp_active = sg_erp[sg_erp['Rates'] >= 0].groupby('RoadName')['Rates'].
mean().reset_index()
69 sg_erp_active.rename(columns={'Rates': 'Avg_ERP_Rate'}, inplace=True)
70
71 sg_model_data = pd.merge(sg_speed, sg_erp_active, on='RoadName', how='inner
').dropna(subset=['SpeedBand', 'Avg_ERP_Rate'])
72 X_sg = sm.add_constant(sg_model_data['Avg_ERP_Rate'])
73 y_sg = sg_model_data['SpeedBand']
74 sg_model = sm.OLS(y_sg, X_sg).fit()
75
76 fig, axes = plt.subplots(1, 2, figsize=(14,6))
77 plt.tight_layout()
78 output_path = Path(__file__).resolve().parent / 'regression.png'
79 plt.savefig(output_path, dpi=160, bbox_inches='tight')

```

Listing 1: experiment_4.py

Experiment: Interactions Among the Three Dimensions

```
1 import pandas as pd
2 import numpy as np
3 import statsmodels.api as sm
4 import matplotlib.pyplot as plt
5 import seaborn as sns
6
7 sns.set_theme(style="whitegrid")
8 plt.rcParams['font.sans-serif'] = ['Arial']
9 plt.rcParams['axes.unicode_minus'] = False
10
11 fig, axes = plt.subplots(1, 2, figsize=(18,8))
12
13 hk_chg = pd.read_csv('Data_5/output/hk_ev_charger_stations.csv')
14 hk_counts = hk_chg.groupby('District').size().reset_index(name='
    Charger_Count')
15
16 hk_inc = pd.read_csv('Data_5/output/hk_income.csv', skiprows=4)
17 hk_inc_2024 = hk_inc[hk_inc['Monthly domestic household income range (3)']
    == '2024'].copy()
18
19 hk_inc_clean = hk_inc_2024[['Unnamed: 1', '>=30,000', 'Total']].copy()
20 hk_inc_clean.columns = ['District', 'High_Income_HH', 'Total_HH']
21 hk_inc_clean['High_Income_Ratio'] = hk_inc_clean['High_Income_HH'] /
    hk_inc_clean['Total_HH']
22
23 hk_model_data = pd.merge(hk_counts, hk_inc_clean, on='District', how='inner
    ')
24 X_hk = sm.add_constant(hk_model_data['High_Income_Ratio'])
25 y_hk = hk_model_data['Charger_Count']
26 hk_model = sm.OLS(y_hk, X_hk).fit()
27
28 sg_inc_raw = pd.read_csv('Data_5/output/sg_income_2015.csv', skiprows=9)
29 sg_inc_raw.columns = sg_inc_raw.iloc[0].values
30 sg_inc = sg_inc_raw.drop(0).dropna(subset=['Planning Area'])
31 sg_inc = sg_inc[~sg_inc['Planning Area'].astype(str).str.startswith('')].
    copy()
32 sg_inc['Planning Area'] = sg_inc['Planning Area'].str.strip()
33
34 cols_to_sum = ['$10,000 - $10,999', '$11,000 - $11,999', '$12,000 - $12,999
    ', '$13,000 - $13,999', '$14,000 - $14,999', '$15,000 - $17,499', '$17
    ,500 - $19,999', '$20,000 & Over']
35
36 for c in ['Total'] + cols_to_sum:
37     sg_inc[c] = pd.to_numeric(sg_inc[c].replace({'-': 0, 'na': np.nan}),
        errors='coerce')
```

```

38
39 sg_inc['High_Income_HH'] = sg_inc[cols_to_sum].sum(axis=1)
40 sg_inc['High_Income_Ratio'] = sg_inc['High_Income_HH'] / sg_inc['Total']
41
42 sg_inc_clean = sg_inc[['Planning Area', 'High_Income_Ratio', 'Total']].
    dropna()
43
44 sg_chg = pd.read_csv('Data_5/output/sg_ev_charger_stations_real.csv')
45 sg_chg['Street Name'] = sg_chg['Street Name'].astype(str)
46 sg_chg['Building Name'] = sg_chg['Building Name'].astype(str)
47
48 def match_sg_area(row, areas):
49     address_text = (str(row['Street Name']) + ' ' + str(row['Building Name']
50     ])).upper()
51     for area in areas:
52         if area.upper() in address_text:
53             return area
54     return 'Unknown'
55
56 sg_areas = sg_inc_clean['Planning Area'].tolist()
57 sg_chg['Mapped_Area'] = sg_chg.apply(lambda row: match_sg_area(row,
58     sg_areas), axis=1)
59
60 sg_counts = sg_chg[sg_chg['Mapped_Area'] != 'Unknown'].groupby('Mapped_Area
61     ').size().reset_index(name='Charger_Count')
62 sg_model_data = pd.merge(sg_counts, sg_inc_clean, left_on='Mapped_Area',
63     right_on='Planning Area')
64
65 X_sg = sm.add_constant(sg_model_data['High_Income_Ratio'])
66 y_sg = sg_model_data['Charger_Count']
67 sg_model = sm.OLS(y_sg, X_sg).fit()
68
69 plt.suptitle('Comparative Analysis of Spatial Equity in EV Infrastructure',
70     fontsize=20, fontweight='bold', y=1.05)
71 plt.tight_layout()
72 plt.savefig('comparative_economy_energy_nexus.png', dpi=300, bbox_inches='
73     tight')

```

Listing 2: experiment_5.py

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AI Declaration

This report has used AI-assisted tools for language refinement, structural editing, and formatting support. All arguments, interpretations, data selection, and final academic judgments are the sole responsibility of the author.